

Exercise 2

Part A : Series Circuit

$$\begin{aligned} 1. R_{\text{total}} &= R_1 + R_2 + R_3 \\ &= 20\Omega + 30\Omega + 50\Omega \\ &= 100\Omega \end{aligned}$$

$$2. I = \frac{V}{R} = \frac{125}{100} = 1.25 \text{ A}$$

$$3. I_1 = I_2 = I_3 = 1.25 \text{ A}$$

$$4. V_1 = 1.25 \text{ A} \times 20\Omega = 25 \text{ V}$$

$$V_2 = 1.25 \text{ A} \times 30\Omega = 37.5 \text{ V}$$

$$V_3 = 1.25 \text{ A} \times 50\Omega = 62.5 \text{ V}$$

$$5. P = V \times I$$

$$P_1 = 25 \text{ V} \times 1.25 \text{ A} = 31.25 \text{ W}$$

$$P_2 = 37.5 \text{ V} \times 1.25 \text{ A} = 46.875 \text{ W}$$

$$P_3 = 62.5 \text{ V} \times 1.25 \text{ A} = 78.125 \text{ W}$$

Parallel Circuit

1. Total Resistance

R_{total}

$$\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$= \frac{1}{20} + \frac{1}{100} + \frac{1}{50} = \frac{5+1+2}{100}$$

$$= \frac{8}{100}$$

$$R_{total} = \frac{100}{8} = 12.5 \Omega$$

$$2 \quad I = \frac{V}{R} = \frac{125V}{12.5 \Omega} = 10A$$

$$3 \quad I_1 = \frac{125V}{20 \Omega} = 6.25A \quad I_3 = \frac{125V}{50 \Omega} = 2.5A$$

$$I_2 = \frac{125V}{100 \Omega} = 1.25A$$

$$4 \quad V_1 = V_2 = V_3 = 125 \text{ V}$$

$$5 \quad P = V \times I$$

$$P_1 = 125 \text{ V} \times 6.25 \text{ A} \\ = 781.25 \text{ W}$$

$$P_2 = 125 \text{ V} \times 1.25 \text{ A} \\ = 156.25 \text{ W}$$

$$P_3 = 125 \text{ V} \times 2.5 \text{ A} \\ = 312.5 \text{ W}$$